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**RAFFLES INSTITUTION**

**2014 Year 6 Preliminary Examination**
Higher 2

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| CANDIDATE NAME | |

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| CIVICS GROUP | 1 | 4 | S | 0 | 3 | | INDEX NUMBER | | | | | | | |

# BIOLOGY
Paper 3

**9648/03**
**22nd SEPTEMBER 2014**
**2 hours**

Additional materials: Answer paper

**READ THESE INSTRUCTIONS FIRST**

Write your index number, CT group & name on all the work you hand in.
Write in dark blue or black pen on both sides of the paper.
You may use a soft pencil for any diagrams, graphs or rough working.
Do not use staples, paper clips, highlighters, glue or correction fluid.

Answer **all** questions.

At the end of the examination, **hand in your planning question (question 4) and essay question (question 5) SEPARATELY.**
The number of marks is given in brackets [ ] at the end of each question or part question.

| For Examiner's Use | |
|---|---|
| **Section A** | |
| **1** | /15 |
| **2** | /13 |
| **3** | /12 |
| **4** | /12 |
| **5** | /20 |
| **Total** | /72 |

Raffles Institution
Internal Examination

This document consists of **13** printed pages and **1** blank page.

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## Section A

Answer **all** the questions in this section.

**1** The human growth hormone (hGH) is a peptide hormone produced by the pituitary gland. It stimulates muscle and bone growth in children.

Fig. 1.1 shows the complete non-template strand of the hGH cDNA in the 5' to 3' orientation.

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  1 TTTCCGACCA TCCCGCTGTC CCGTCTGCAG GATAACGCTA TGCTGCGTGC ACACCGCCTG
 61 ---------- ---------- ---------- ---------- ---------- ----------
121 ---------- ---------- ---------- ---------- ---------- ----------
181 ---------- ---------- ---------- ---------- ---------- ----------
241 ---------- ---------- ---------- ---------- ---------- ----------
301 ---------- ---------- ---------- ---------- ---------- ----------
361 ---------- ---------- ---------- ---------- ---------- ----------
421 ---------- ---------- ---------- ---------- ---------- ----------
481 ---------- ---------- ---------- ---------- ---------- ----------
541 CAGTGTCGTT CCGTGGAGGG TTCCTGTGGC TTCTTTTGAG GTGTGAGCTC TAATAGTAAT
```

**Fig. 1.1**

**(a)** The coding sequence can be found on the non-template strand of the hGH cDNA. Circle the first and the last codon of the coding sequence of the hGH on Fig. 1.1. [2]

**(b)** Describe how the cDNA shown in Fig. 1.1 is produced.

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**(c)** Describe 2 ways in which cDNA differs from genomic DNA.

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**(d)** The hGH cDNA is present in low copy number in the cDNA library. PCR can be used to amplify the hGH cDNA so that it can be used subsequently for genetic engineering.

With reference to Fig 1.1, suggest the sequence of two 10-nucleotide long primers that can be used to amplify the complete hGH cDNA.

1.…………...……………………………...…………………………………………………

2 ………..………………………………………………………………………………….[2]

hGH is frequently administrated during hormone replacement therapy to patients with growth hormone deficiencies. In the past, the hGH used for treatment was harvested from cadavers. The hGH used currently is produced using recombinant DNA technology.

Fig.1.2 shows a diagram of the plasmid pUC18 and the position of the restriction sites found in the multiple cloning site (MCS).

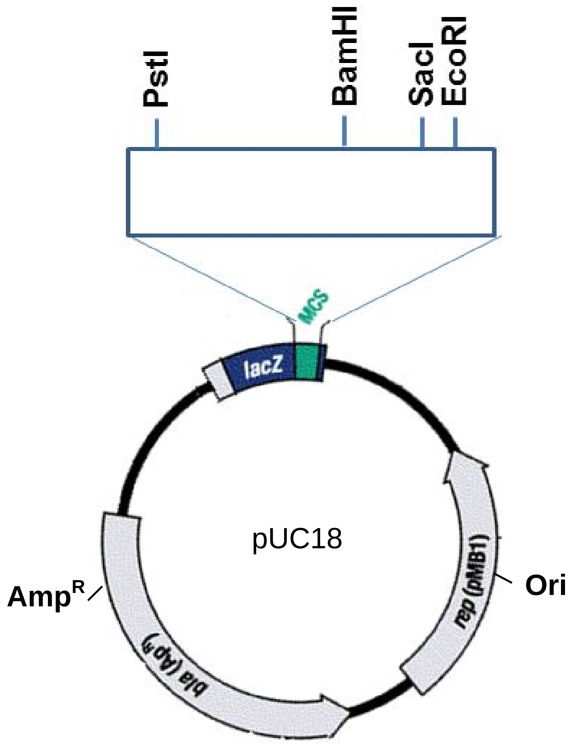



**Fig. 1.2**

Table 1.1 shows some details of four restriction enzymes frequently used in genetic engineering.

| name | site |
|---|---|
| *Bam*H1 | 5'... G↓GATCC…3'<br>3'…CCTAG↑G…5' |
| *Eco*RI | 5'... G↓AATTC…3'<br>3'…CTTAA↑G…5' |
| *Pst*I | 5'... CTGCA↓G…3'<br>3'…G↑ACGTC…5' |
| *Sac*I | 5'... GAGCT↓C…3'<br>3'…C↑TCGAG…5' |

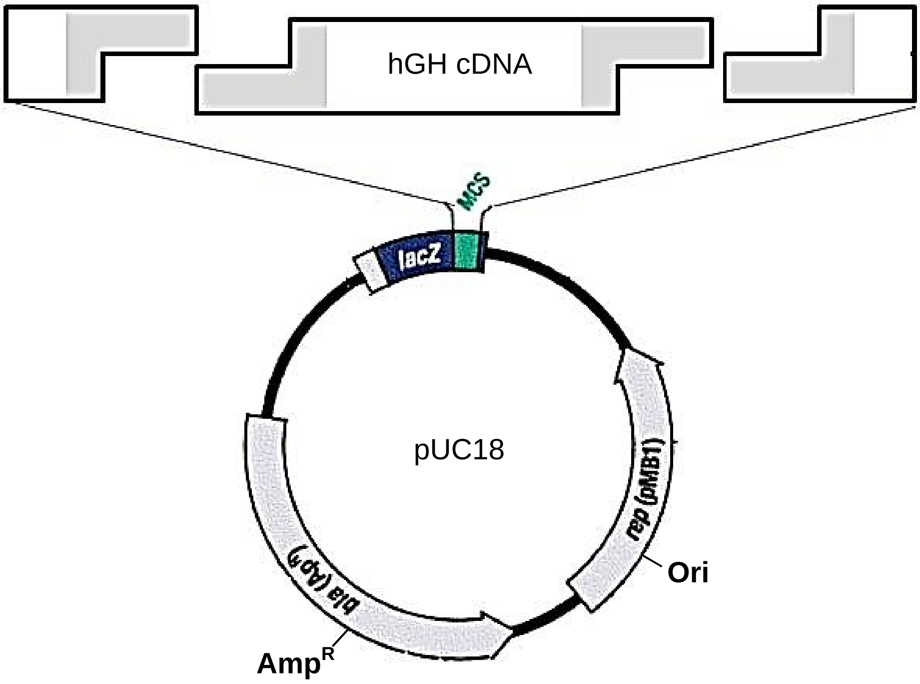
**Table 1.1**

**(e)** The recombinant DNA used to produce hGH is produced by cutting the cDNA (Fig. 1.1) with two different restriction enzymes. The same two restriction enzymes are also used to cut the plasmid. This generates two different types of sticky ends on both the plasmid and cDNA which enables directional cloning where the cDNA is inserted into pUC18 in only one possible orientation.

**(i)** With reference to the position (Fig. 1.2) and details (Table 1.1) of the restriction sites on the plasmid study the sequence of the hGH cDNA (Fig. 1.1) and identify the two restriction enzymes that should be used to produce the recombinant plasmid.

…………..…………………………………………………………………………………….[2]

**(ii)** The diagram below shows how the sticky ends generated by the two restriction enzymes identified in **d(i)** are able to anneal to form the recombinant plasmid. Complete the diagram by writing the nucleotide sequence of the sticky ends in the shaded boxes provided.




[2]

**(iii)** Name one other enzyme that is used in the formation of this recombinant plasmid and describe its function.

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[Total:15 marks]

**2** When police investigators arrived at a crime scene, a woman was found dead in a pool of her blood. Her younger brother was sitting beside her, holding a blood stained knife. He is a 15 year old boy with Down's syndrome and could not help in the investigation due to his irregular speech. Investigators detained the victim's boyfriend and 2 other suspects.

As part of the forensic analysis, the D21S19 locus, found on chromosome 21, which contains a variable number of tandem repeats (VNTRs) was examined. Polymerase chain reaction (PCR) was used to amplify the D21S19 locus from the various DNA samples obtained from the scene. This was followed by separation of the DNA fragments using gel electrophoresis.

Fig. 2.1 shows the results following gel electrophoresis.

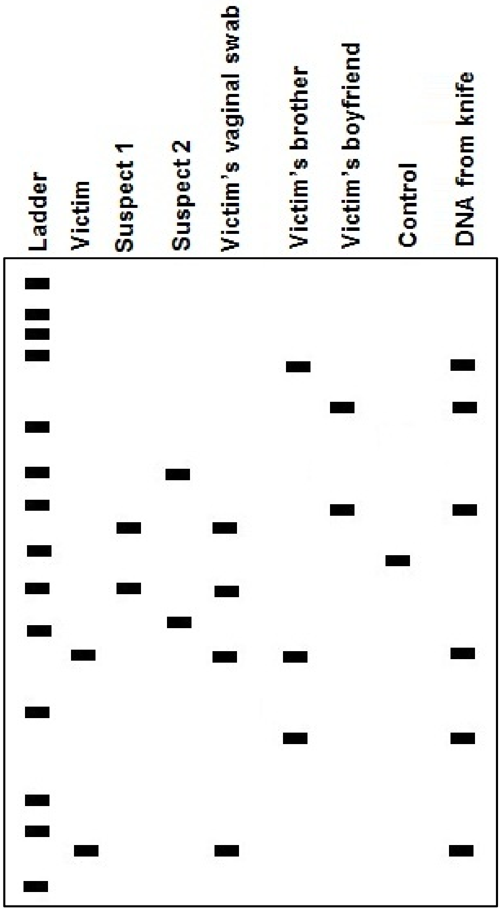


**Fig. 2.1**

**(a)** Taq polymerase instead of human DNA polymerase is used in the PCR. Explain why Taq polymerase is used.

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……………………………………………………………………………………………………[1]

**(b)** DNA containing the D21S19 locus with a known number of tandem repeats was used as

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a control when performing the PCR. Explain the purpose of this control.

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…………………………………………………………………………………………………...[1]

**(c)** Explain the banding pattern seen in the analysis of the victim's brother's DNA.

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**(d)** With reference to Fig. 2.1,

**(i)** identify the rapist;

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**(ii)** provide an explanation for your answer in **d (i)**.

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**(e)** The above analysis used a single VNTR locus. Explain why DNA profiling labs normally compare 13 such loci when conducting investigations.

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**(f)** Explain why structural genes are seldom used for genetic fingerprinting.

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…………………………………………………………………………………………………..[2]

[Total: 13 marks]

**3** Fig. 3.1 below provides information on genetically engineered (GE) Atlantic salmon.

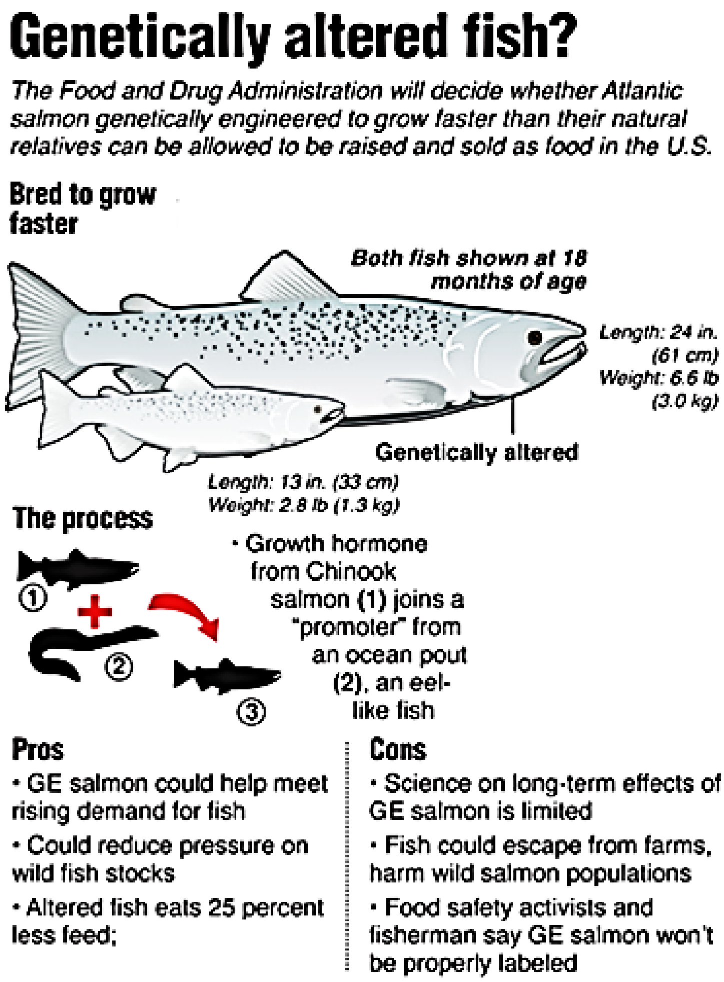


**Fig. 3.1**

**(a)** With reference to Fig. 3.1, explain how farmers can benefit from farming GE salmon as compared to farming wild salmon.

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…………………………………………………………………………………………………..[2]

**(b)** **(i)** Name the gene from which the ocean pout promoter was derived.

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……………………………………………………………………………………………………[1]

**(ii)** Explain the significance of using this promoter in the GE salmon.

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……………………………………………………………………………………………………[1]

**(c)** State one precaution which has been taken to reduce the risk of the GE salmon escaping from the farms and harming wild salmon populations.

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……………………………………………………………………………………………………[1]

Fig. 3.2 shows organogenesis of tobacco plant calli incubated on nutrient media containing different concentrations of IAA (an auxin) and kinetin (a cytokinin).

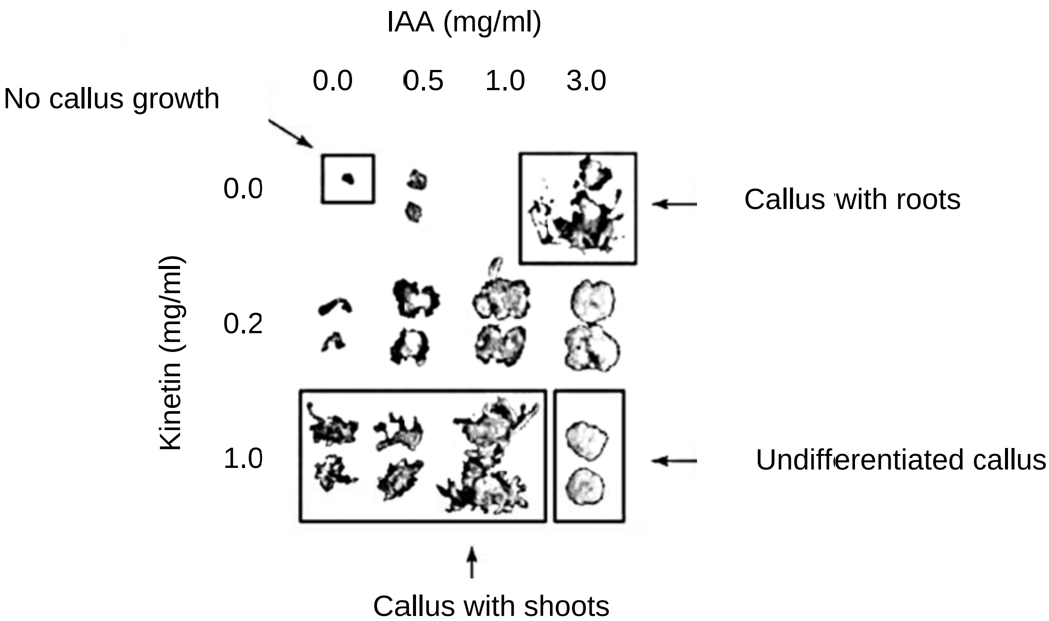


**Fig. 3.2**

**(d)** Using information from **Fig. 3.2**, describe how to obtain virus-free callus that is genetically identical to a diseased tobacco plant.

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……………………………………………………………………………………………………[2]

**(e)** In nature, *Agrobacterium tumefaciens*, a soil bacterium, induces crown gall tumours in the roots of leguminous plants. The bacterium integrates its T-DNA (as shown in Fig. 3.3), found in the Ti plasmid, into the plant genome. Expression of this group of genes found on the T-DNA is necessary for tumour formation.

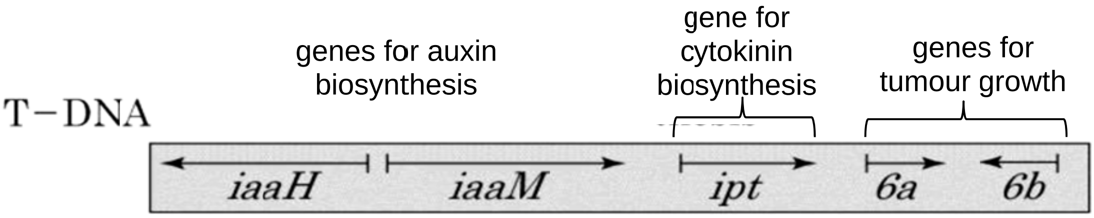


**Fig 3.3**

Various mutations were introduced to the T-DNA in the Ti plasmid and the details are shown in the table 3.1 below.

| **Ti plasmid** | **mutations on T-DNA** |
|---|---|
| Ti plasmid I | deletion of *iaaH & iaaM*, gain-in-function mutation of *ipt* |
| Ti plasmid II | gain-in-function mutation of *iaaH & iaaM*, deletion of *ipt* |
| Ti plasmid III | deletion of *iaaH, iaaM* and *ipt* |

**Table 3.1**

3 different *Agrobacterium* samples were transformed with Ti plasmids I, II and III. The 3 transformed *Agrobacterium* samples were then used to infect different calli. The resultant transformed calli were then grown on media deficient in plant growth regulators.

Using information from Fig 3.2 and Fig 3.3, predict what will happen to the resultant callus.

| ***Agrobacterium* transformed with** | **effect on resultant transformed calli** |
|---|---|
| Ti plasmid I | |
| Ti plasmid II | |
| Ti plasmid III | |

[3]

**(f)** *Agrobacterium*-mediated transformation involves the removal of tumour-inducing genes and the genes involved in auxin and cytokinin biosynthesis and the insertion of the gene of interest (e.g. Bt toxin gene) fused with a kanamycin resistant gene into the T-DNA. Explain how the transformed plant cells are identified.

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……………………………………………………………………………………………[2]

[Total: 12 marks]

**4 Planning question**

Raw pineapple is a rich source of trypsin, a protease.

Fig 5.1 shows a section of a piece of photographic film. The film is made up of a transparent plastic backing coated with a dark layer that is made up of a mixture of gelatin (protein) and silver halides. The gelatin functions as a binding medium where silver halides are embedded.

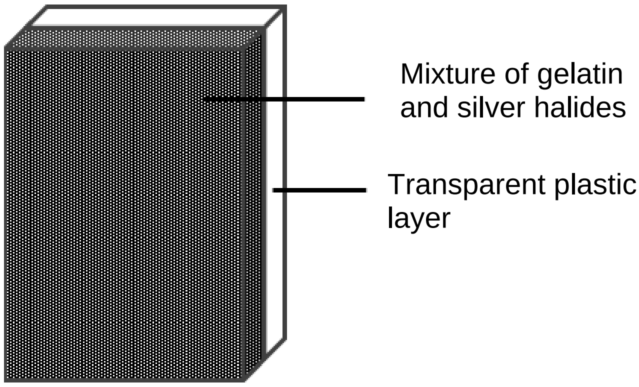


**Fig. 5.1**

Treatment of photographic film with trypsin will result in the digestion of the gelatin layer, and the release of the silver halides embedded in it.

Upon complete digestion, the dark layer can be easily removed by rubbing the film with your fingers. This will reveal the transparent plastic backing.

Using this information and your own knowledge, design an experiment to test the hypothesis that:

**"The rate of trypsin activity is dependent on the pH of the solution."**

You must use the following:

- pineapple fruit
- food blender
- muslin cloth
- pH buffers (choose from range of 3-11)
- distilled water
- a piece of photographic film (3cm by 200cm)
- forceps

You may select from the range of common laboratory apparatus e.g.:

- any normal laboratory glassware e.g. test-tubes, beakers, measuring cylinders, graduated pipettes, glass rods, etc.,
- Bunsen burner
- thermometer
- syringes
- timer e.g. stopwatch or stopclock
- etc,

Your plan should have a clear and helpful structure to include:

- a description of the method used including the scientific reasoning behind the method;
- an annotated diagram, if necessary;
- an explanation of the dependent and independent variables involved;
- how you will record your results and ensure they are as accurate and reliable as possible;
- proposed layout of results tables and graphs with clear headings and labels;
- the correct use of technical and scientific terms and
- relevant risks and precautions taken.

[Total: 12 marks]

**5** **Free-response question**

Write your answers to this question on the separate answer paper provided.
Your answers:

- should be illustrated by large, clearly labelled diagrams, where appropriate,
- must be in continuous prose, where appropriate,
- must be set out in sections (a), (b) etc. as indicated in the question.

**(a)** The two common forms of severe combined immunodeficiency (SCID) are adenosine deaminase (ADA) deficiency and X-linked SCID. Compare the differences between these two forms of SCID and describe how viral-mediated gene therapy can be used to treat SCID. [8]

**(b)** Explain how nucleic acid hybridization and RFLP analysis can be used to detect a named disease. [6]

**(c)** Discuss the ethical concerns that have arisen regarding the human genome project. [6]

[Total: 20 marks]

**End of Paper**